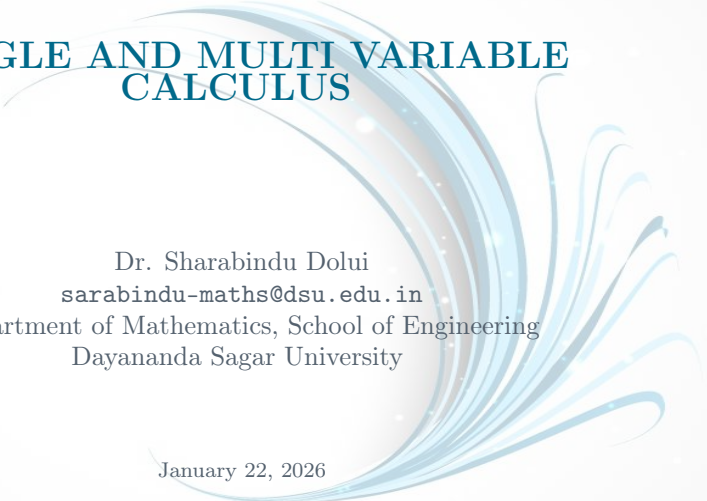


SINGLE AND MULTI VARIABLE CALCULUS



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Course Syllabus



- ▶ **UNIT I: Differential Calculus** Functions of two or more variables, Regions in the plane, Level curves and level surfaces, Limits and continuity, Partial derivatives and differentiability, Extreme values, saddle points, Lagrange multipliers.
- ▶ **UNIT II: Integral Calculus** Double integrals and iterated integrals, Cartesian and polar coordinates, Triple integrals, Change of variables, Multiple integrals in cylindrical and spherical coordinates.
- ▶ **UNIT III: Vector Calculus** Vector fields and line integrals, Work, circulation, and flux, Path independence and conservative fields, Green's theorem, Surface area and surface integrals, Parametrized surfaces, Stokes' theorem and Divergence theorem.
- ▶ **UNIT IV: Sequence and Series I:** Sequences of real numbers and their convergence criteria, Infinite series, Sequence of partial sums, Tests for convergence/divergence - nth term test, Boundedness and monotonicity, Integral, Condensation, Comparison, Ratio and root tests.
- ▶ **UNIT V: Sequence And Series II:** Alternating series, Absolute and conditional convergence, Rearrangement theorem, Power series, Taylor and Maclaurin series (one and two variables).



Textbooks

1. George B. Thomas, M. D. Weir, and J. Hass, *Thomas' Calculus*, 13th Edition, Pearson, 2014.
2. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley India, 2015.

Reference Books

1. B. S. Grewal, *Higher Engineering Mathematics*, 43rd Edition, Khanna Publishers, 2015.
2. John Bird, *Higher Engineering Mathematics*, 6th Edition, Elsevier, 2017.
3. James Stewart, *Calculus: Early Transcendentals*, 8th Edition, Cengage Learning, 2017.
4. K. A. Stroud and Dexter J. Booth, *Engineering Mathematics*, 7th Edition, Palgrave Macmillan, 2013.
5. J. E. Marsden, A. J. Tromba, and A. Weinstein, *Basic Multivariable Calculus*, 3rd Edition, W. H. Freeman.

E-Resources

- ▶ Youtube- Multivariable Calculus by Dr. S. K. Gupta
- ▶ Youtube- Infinite Series (Convergence & Divergence) by GP Sir



- ▶ What is a real-valued function $f(x)$?
- ▶ What are the domain and range of a function?
- ▶ What is a limit?
- ▶ What is continuity?
- ▶ What is differentiation?
- ▶ How to find local and global extreme values of a function?

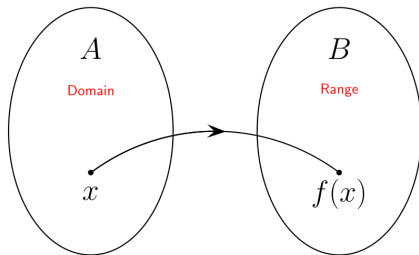
Function



A **function** from a set A to a set B assigns to each element of A exactly one element of B . The set A is called the domain of the function and the set B is called the codomain of the function.

$$f : A \rightarrow B,$$

For a **real valued function**, $A, B \subseteq \mathbb{R}$ and $f(x) \in \mathbb{R}$ for all $x \in A$.





Find the **domain** and **range** of the following functions:

► $y = x^2$

► $y = \frac{1}{x}$

► $y = \sqrt{x}$

► $y = \sqrt{4 - x}$

► $y = \sqrt{1 - x^2}$

► $y = e^x$

► $y = \sin x$

► $y = \tan x$

Also draw their graphs

Limit of a Function



Let f be a real-valued function defined on a set $D \subseteq \mathbb{R}$, and let $a \in \mathbb{R}$ be a limit point of D . We say that $b \in \mathbb{R}$ is a **limit of $f(x)$ as x approaches a** if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - b| < \varepsilon \quad \text{whenever } x \in D \text{ and } 0 < |x - a| < \delta. \quad (*)$$

In that case, we write

$$\lim_{x \rightarrow a} f(x) = b,$$

or equivalently,

$$f(x) \rightarrow b \quad \text{as } x \rightarrow a.$$

The relation in $(*)$ can also be written as

$$x \in D, \quad 0 < |x - a| < \delta \Rightarrow |f(x) - b| < \varepsilon.$$

- ▶ In the following, whenever we talk about the limit of a function f as x approaches $a \in \mathbb{R}$, we assume that f is defined on a set $D \subset \mathbb{R}$ and that a is a limit point of D .
- ▶ Show that a function cannot have more than one limit.



Find the limits of the following functions, if they exist:

► $\lim_{x \rightarrow 4} \sqrt{x^2 - 10}$

► $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$

► $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$

► $\lim_{x \rightarrow 0} \frac{|x|}{x}$

► Find the limit of the function $f(x)$ as $x \rightarrow 0$, where

$$f(x) = \begin{cases} 1, & x \leq 0, \\ \sin x, & x > 0. \end{cases}$$

Continuity



Let f be a real-valued function defined on an interval I .

1. The function f is said to be **continuous at** $a \in I$ if

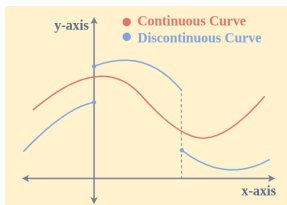
$$\lim_{x \rightarrow a} f(x) = f(a).$$

2. The function f is said to be **continuous on** I if f is continuous at every $a \in I$.

Using the definition of limit, it follows that

- f is continuous at $a \in I$ if and only if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - f(a)| < \varepsilon \quad \text{whenever } x \in I \text{ and } |x - a| < \delta.$$





1) For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous at every x ?

2) Find all values of a such that the function

$$f(x) = \begin{cases} x^2 + 2x, & x < a, \\ -1, & x \geq a, \end{cases}$$

is continuous everywhere.



Suppose f is a (real-valued) function defined on an open interval I and $x_0 \in I$. Then f is said to be **differentiable at** x_0 if

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists, and in that case the value of the limit is called the **derivative** of f at x_0 . The derivative of f at x_0 , if it exists, is denoted by

$$f'(x_0) \quad \text{or} \quad \frac{df}{dx}(x_0).$$

Differentiability implies continuity: Suppose f defined on I is differentiable at $x_0 \in I$. Then f is continuous at x_0 .

A function can be **continuous but not differentiable**. The function $|x|$ is continuous everywhere but not differentiable at $x = 0$.



Find the **Derivatives** of the following functions:

- ▶ $f(x) = x^n$
- ▶ $y = e^x$
- ▶ $y = \frac{1}{x}$
- ▶ $y = \sqrt{x}$
- ▶ $y = \sin x$
- ▶ $y = \tan x$
- ▶ $y = e^{ax} \cos(bx)$

Also find the **Integration** of these functions.